北京郵電大學

Queen Mary University of London

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# EBU6501 A

**Joint Programme Examinations 2022/23**

**EBU6501 Middleware**

**Paper A**

**Time allowed 2 hours**

**Answer ALL questions**

**Complete the information below about yourself very carefully.**

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| **QM student number** | | | | | | | | | | |
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**NOT allowed: Electronic calculators and electronic dictionaries.**

**INSTRUCTIONS**

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

**Examiners**

Dr Gokop Goteng, Dr Ethan Lau

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Filename: 2223_EBU6501_A No answer book required

# Instructions

**Before the start of the examination**

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

**During the examination**

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

**At the end of the examination**

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

**Question 1**

a) Maintaining the order, explain how the communication of messages within the Message-oriented middleware (MOM) is achieved within the objects **`MessageProducer`** and **`MessageConsumer`**. Draw the MOM architecture diagram to aid the explanation.

**(5 marks)**

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| | **5 marks** |

b) Apache ActiveMQ is an example of the Message Broker for the Java Message Service (JMS) application. It is used in publishing messages to a ***Topic*** or ***Queue***.

**(8 marks)**

i) The Figure 1(a) and 1(b) show the type of the JMS messaging model. Denote the type of the messaging model used in a) and b). Then, explain the difference between the models.

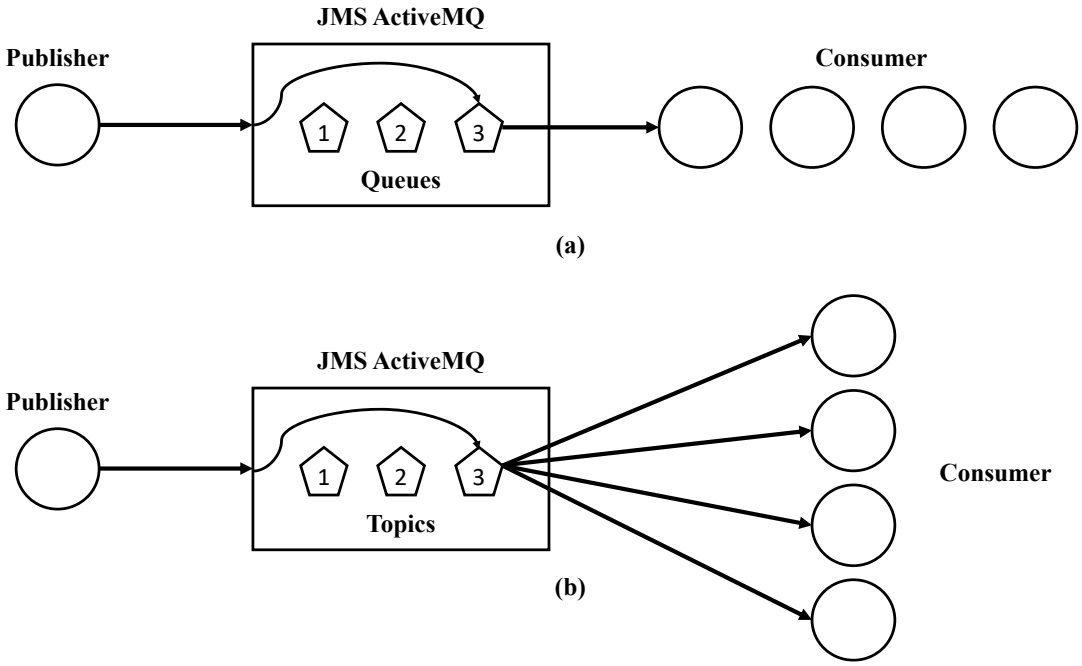



**Figure 1 (a-b).**

**[5 marks]**

ii) In your own words, describe the importance of the ***Sessions*** in the context of JMS Application Programming Interface (API) model.

**[3 marks]**

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| | | **8 marks** |

c) Figure 2 below shows the directory structure of the Tomcat container.

**(5 marks)**

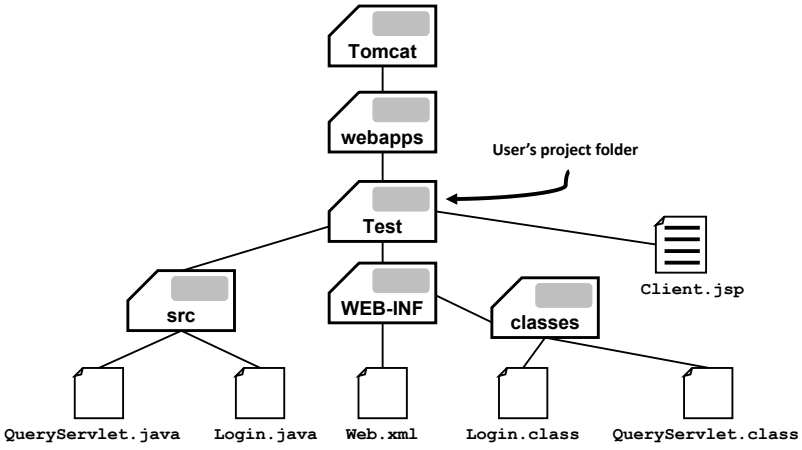



**Figure 2.**

i) Carefully denote the pages that are belong to the individual design pattern of the ***Model***, ***View*** and ***Controller*** model.

**[3 marks]**

ii) Define the term given to the file "**web.xml**" and explain why it is located under the **WEB-INF** directory**.**

**[2 marks]**

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d) Figure 3 presents the usage of Java Servlet implementation of the ***Request-Response*** programming model which responses to client's request by querying the database. The **`QueryServlet`** is created that extends **`HttpServlet`** class. The code requires the completion of the **`Service`** and **`Destroy`** methods. The servlet shall return an HTML page that display the query response the string field **`myStatus = "<Query Response>"`**. Complete the following:

**(7 marks)**

```
public class QueryServlet extends HttpServlet {

      // ### Missing the service() method

      // ### Missing the destroy() method
                        }
```

**Figure 3.**

i) Write the required ***`service()`*** method that called by the server to allow a servlet to handle a GET request.

**[2 marks]**

ii) Set the content type as "text/html" for the webpage.

**[1 mark]**

iii) Write message ***`myStatus`*** to the webpage.

**[2 marks]**

iv) Close the stream for garbage collection and destroy the servlet.

**[2 marks]**

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| | | **7 marks** |

**Question marking**

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**Question 2**

a) The Inversion of Control (IoC) is a principle in the software engineering of the Spring Framework as shown in Figure 4. It is most often used in the context of object-oriented programming such as Java.

**[10 marks]**

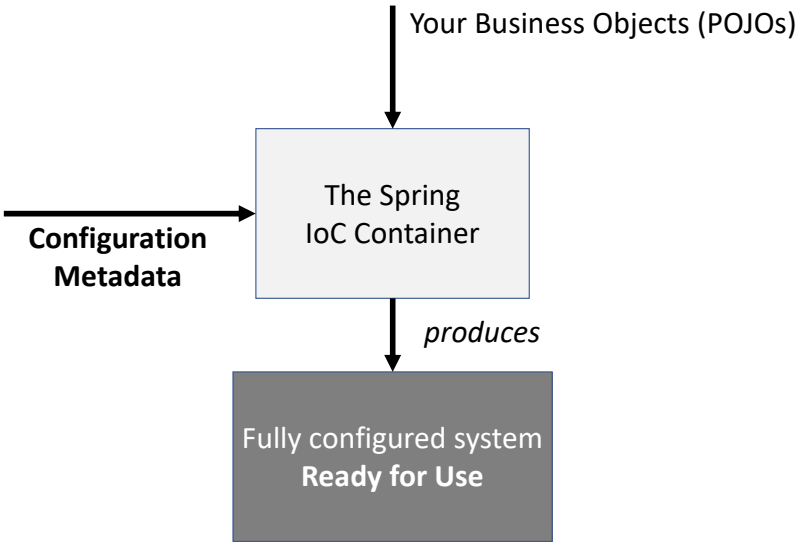



**Figure 4**

i) Based on the information in Figure 4, provide TWO types of the spring IoC container and further explain its role in the Spring Framework.

**(4 marks)**

ii) List out the TWO examples of the configuration metadata.

**(2 marks)**

iii) The Application Context is Spring's advanced container. Implement the Spring's Application Context by instantiating the **`ApplicationContext`** interface that loads the .xml configurations from a classpath of "**info.xml**". Finally, obtain the bean with the bean ID = **`beanInfo`** from the **`beanInfo`** class.

**(4 marks)**

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| | | **10 marks** |

b) This question is about Java threads.

**[15 marks]**

i) Maintaining their order, state and explain the main FIVE life cycle states of a Java thread.

**[6 marks]**

ii) In your own words, explain the ***Visibility*** and ***Race Conditions*** concurrency issues.

**[4 marks]**

iii) Suggest the corresponding methods to avoid the ***Visibility*** and ***Race Conditions*** issues.

**[2 marks]**

iv) Figure 5 shows the Java class **`MyAccumulator`** where the method **`getNextValue()`** is not thread-safe as it is accessible by multiple threads. Improve the code to ensure the thread-safe and atomic operation.

```
public class MyAccumulator {
  private int value = 500;
  public int getNextValue(){
        return value++;
  }
}
```

**Figure 5.**

**[3 marks]**

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| | | **15 marks** |

**Question marking**$= \frac{}{10} + \frac{}{15} = \frac{}{25}$

**Question 3**

a) An Internet of Things (IoT) company is using thousands of sensors to receive, process and send data in form of messages to heterogeneous distributed computer platforms across many countries as logs. The data come from many sources to forecast weather, mortgage rates, inflation, currency exchange rates and many others. The system that stores this data as logs is not well organised and it is difficult to identify and separate the data into different areas (weather, mortgage, inflation, etc) to make any meaning from them. A Middleware Developer is hired by the company to develop a system that will ensure data is well organised and durable using Apache Kafka.

**[18 marks]**

i) Describe how the Middleware Developer will ensure that the different data from the different sources will be well organised (weather, mortgage rates, etc should be separate) and durable (last for a long time) using Kafka.

**(8 marks)**

ii) Describe the TWO mechanisms that the Middleware Developer can use to manage and convert the internal data (machine readable) to external data (human readable) formats when data is transported from sensors to output units for users to use and vice versa.

**(2 marks)**

iii) If the Middleware Developer has written the Producer Library for a mortgage rate payment as shown in Figure 6 below, write the corresponding Consumer Library using the Java class KafkaConsumer and Java collection ConsumerRecords in Apache Kafka.

**(8 marks)**

```
try (KafkaProducer<String, Payment> producer = new KafkaProducer<>(props)) {

    for (long i = 0; i < 10; i++) {
        final String custId = "id" + Long.toString(i);
        final Payment payment = new Payment(custId, 1000.00);
        final ProducerRecord<String, Payment> record =
           new ProducerRecord<>("transactions",
                                    payment.getId().toString(),
                                    payment);
        producer.send(record);
   }
} catch (final InterruptedException e) {
    e.printStackTrace();
}
```

**Figure 6: Kafka Producer Library**

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| | **18 marks** |

b) Describe TWO methods in which you can use **Cryptography** to ensure secure sharing of files using one key and two keys respectively in a computer network.

**[5 marks]**

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| | **5 marks** |

c) Describe ONE IoT protocol that is the most commonly used in homes and is suitable for short and medium range distance coverage in a Local Area Network (LAN).

**[2 marks]**

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| | **2 marks** |

**Question marking:** $\frac{}{18} + \frac{}{5} + \frac{}{2} = \frac{}{25}$

**Question 4**

a) Using your knowledge of the Open Services Gateway Initiatives (OSGi) architecture, describe how ***Bundles***, ***Security*** and ***Services*** work in OSGi.

**[6 marks]**

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| | **6 marks** |

b) A company has been using physical machines to run its software and hardware middleware systems. The company wants to move all its IT infrastructure to the cloud and is looking for a way of eliminating the time and effort of provisioning compute and storage servers. It also wants to adopt the methodology of running and processing data in automated small pieces of actions that are faster and more efficient. Describe TWO types of services that can achieve these requirements for the company using TWO examples (ONE for each service) of such services.

**[10 marks]**

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c) Describe the FIVE main steps in using a Uniform Resource Locator (URL) in a networking Java program (use Java classes and methods where possible).

**[9 marks]**

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